PROJECT GUIDE · DATA

StreetCare AI CSV Schemas

The schema of every CSV file in the project: every column with its meaning, its type after cleaning, how empty it is, its range or values, an example as written, its role in the models and the exact rule that cleans it — with row counts, keys, targets and how the files relate.

13 CSV files | 8 schemas | 163 column definitions | 3 targets | Relationship diagram | Measured on a fresh build

Wrenford Borough Council · Street Care Office · figures from data/*.csv, processed/*.csv and output/data_quality_report.json

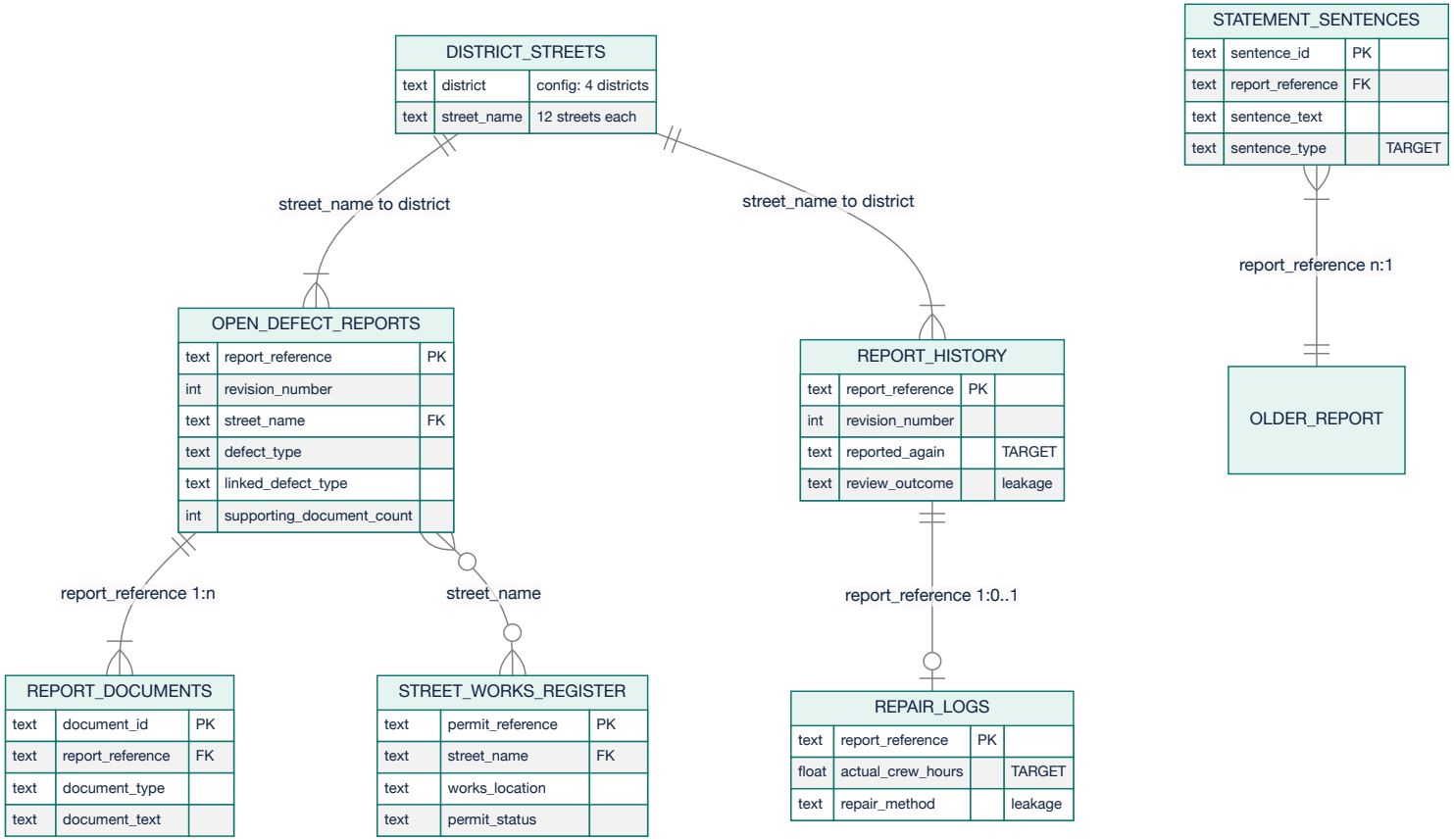
All 13 CSV files

Layer	File	Rows	Cols	Primary key	Target	Used for
Training extract	`data/report_history_train.csv`	32,738	31	report_reference (+ revision_number before cleaning)	reported_again	Trains the repeat likelihood model; counted for chat street figures.
Training extract	`data/report_history_eval.csv`	8,202	31	report_reference (+ revision_number before cleaning)	reported_again	Scores the repeat likelihood model; chat street figures.
Training extract	`data/repair_logs_train.csv`	26,095	27	report_reference (one log per report; a repeat is refused)	actual_crew_hours	Trains the expected crew hours model.
Training extract	`data/repair_logs_eval.csv`	6,476	27	report_reference	actual_crew_hours	Scores the expected crew hours model.
Training extract	`data/statement_sentences.csv`	36,661	8	sentence_id	sentence_type	Trains the sentence type model (stratified 80/20 split, seed 1313).
Live input	`data/input/open_defect_reports.csv`	45	27	report_reference (+ revision_number before cleaning)	—	The queue, screening, both models at prediction time, the review workflow.
Live input	`data/input/report_documents.csv`	62	6	document_id; report_reference → open_defect_reports	—	Statement evidence: sentence model and disagreement flags.
Live input	`data/input/street_works_register.csv`	29	6	permit_reference; street_name → open_defect_reports	—	Live and conflicting permit counts, the live street works trigger.
Cleaned output	`processed/report_history_train_clean.csv`	30,569	31	as raw	reported_again	written by run_data_pipeline.py; read by model training
Cleaned output	`processed/report_history_eval_clean.csv`	7,648	31	as raw	reported_again	written by run_data_pipeline.py; read by model training
Cleaned output	`processed/repair_logs_train_clean.csv`	25,537	27	as raw	actual_crew_hours	written by run_data_pipeline.py; read by model training
Cleaned output	`processed/repair_logs_eval_clean.csv`	6,337	27	as raw	actual_crew_hours	written by run_data_pipeline.py; read by model training

Cleaned output	`processed/statement_sentences_clean.csv`	35,717	8	as raw	sentence_type	written by run_data_pipeline.py; read by model training

The three live files are not written back: they are cleaned in memory each time the service loads them (45 → 40 reports, 62 → 60 documents, 29 → 28 permits). Each cleaned file has the same columns as its raw file; only values, types and rows change.

How the files relate



Link	Cardinality	What was measured
report_history_train → report_history_eval report_reference	disjoint	train and evaluation extracts never share a report (no leakage). **0 shared references**
repair_logs_train → repair_logs_eval report_reference	disjoint	train and evaluation repair logs never share a report. **0 shared references**

report_history_* → repair_logs_* report_reference	one decided report : 0..1 repair log	a repair log records the repair of a decided report; each model still trains only on its own extract. **31644 of 31874 log references are in the 38217 history references; by split: train history × train logs 25352, train history × eval logs 0, eval history × train logs 0, eval history × eval logs 6292**
statement_sentences → report_history_* report_reference	many sentences : one report	labelled sentences of documents attached to older reports. **5902 reports, median 6 sentences per report; 0 of them in the history extracts**
report_history_* (raw) → report_history_* (clean) report_reference + revision_number	many revisions : one kept row	a report resubmitted keeps only its highest revision. **duplicates and older revisions removed in cleaning**
open_defect_reports → report_documents report_reference	one report : 0..n documents	documents attached to each open report (statement evidence). **60 documents for 40 of 40 reports; 0 reports have none; 0 documents point at no open report; supporting_document_count matches in 40 of 40**
open_defect_reports → street_works_register street_name	many reports : many permits (via street)	live permits on the report's street give live_street_works_on_street; those whose works_location is the report's highway part are conflicting. **28 permits (23 Live) on 26 streets; 19 of 40 reports are on a street with a live permit**
open_defect_reports → report_history_* report_reference	disjoint	the live queue holds new reports, never ones already decided. **0 shared references**
report_history_train → config.INSPECTION_DISTRICT_STREETS street_name → district	many streets : one district (48 streets, 4 districts × 12)	every street belongs to exactly one inspection district. **0 rows whose district disagrees with the street's**
repair_logs_train → config.INSPECTION_DISTRICT_STREETS street_name → district	many streets : one district (48 streets, 4 districts × 12)	every street belongs to exactly one inspection district. **0 rows whose district disagrees with the street's**
open_defect_reports → user_accounts (Highways Inspector) street_name → district → inspector.district	one district : 2 inspectors	only an inspector of the street's district may open or decide its review (authorize_district). **0 live reports whose district disagrees with the street's**

Conventions used in every schema

Everything is read as text first

Every raw CSV is read with `dtype=str`, each cell trimmed. The **Type** column of each schema is the type *after* cleaning: text, whole number (Int64, allows blanks) or decimal.

What counts as empty

A blank cell or any null token, case-insensitive: `n/a` `na` `–` `null` `not known` `caller unsure`. **Null %** is measured after cleaning.

Report references

`SC0052125`, `SC-52125` and `52125` are one report: all become SC + 7 zero-padded digits.

Readings with units

mm / cm / in → mm (whole); m / ft → m (0.1); hrs / min → hours (0.1); deg → degrees (0.1). Exact Decimal maths, rounded half-up. Values below the minimum (or lean above 90°) become blank.

Categories

Every spelling is folded onto one canonical value through a `config.*_MAPPING` (227 spellings in all), e.g. `cw`, `road surface` → Carriageway.

Rows that are removed or refused

Exact duplicate rows are removed; earlier revisions of a report are removed; rows with no target are removed. A repeated identifier with different content is **refused** (ValueError), never guessed.

Role badges

TARGET what a model predicts · FEATURE a model input · KEY identifier / join column · LEAKAGE known only after the decision, so never a model input · context kept for screening, documents or display.

CSV SCHEMA · data/report_history_train.csv

Report history · train

Decided defect reports, one row per revision, with whether the defect came back.

32,738 → 30,569
rows raw → clean

2,169
removed: 369 duplicates, 1,565 earlier revisions, 235 no target

31
columns

reported_again
target

report_reference (+ revision_number before cleaning)
primary key

Cleaned file: `processed/report_history_train_clean.csv` · **Used for:** Trains the repeat likelihood model; counted for chat street figures.

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
1	**report_reference** Report identifier, canonical SC + 7 digits	text	0.0%	30,569	30,569 distinct	`SC-24433 | SC0020778 | SC-16501`	KEY	converted to canonical SC + 7 digits (REPORT_REFERENCE_PATTERN: SC0052125 / SC-52125 / 52125)
2	**report_channel** How the report came in	text	0.0%	3	Website (12882) StreetCare app (11570) Contact centre (6117)	`StreetCare app | Website | Contact centre`	context	mapped to 3 canonical values via REPORT_CHANNEL_MAPPING (3 variants)
3	**revision_number** Revision of the report; only the latest is kept	whole number	0.0%	3	1 – 3 median 1.0	`1 | 2 | 3`	context	whole number via POSITIVE_WHOLE_NUMBER_PATTERN, Int64; only the highest revision of each report is kept
4	**reporter_type** Who reported it	text	0.91%	5	Resident (25028) Business (2122) Councillor (1434) School (1020) Bus operator (688)	`Resident | Councillor | School`	context	mapped to 5 canonical values via REPORTER_TYPE_MAPPING (5 variants)
5	**street_name** Street of the defect (one of 48)	text	0.0%	48	48 distinct	`Holloway Lane | Long Lane | Pond Lane`	context	mapped to 48 canonical values via STREET_NAME_MAPPING (48 variants)
6	**district** Inspection district (one of 4)	text	0.0%	4	Town Centre (9009) North (8137) South (7977) Villages (5446)	`Villages | villages | Town Centre`	context	mapped to 4 canonical values via DISTRICT_MAPPING (4 variants)
7	**street_category** Street category (e.g. Main route, Local road)	text	1.34%	2	Local road (16301) Traffic sensitive (13857)	`Residential street | Traffic sensitive | Local road`	FEATURE	mapped to 2 canonical values via STREET_CATEGORY_MAPPING (6 variants)

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
8	**location_description** Free-text location	text	0.0%	479	479 distinct	`by lamp column 44 | at the junction with Clay Pit Lane | at the junction with Littleworth`	context	trimmed; null tokens → blank; otherwise kept as written
9	**location_fixed** Location fixed to a house no., lamp column or junction	text	1.23%	2	Yes (26418) No (3774)	`true | Yes | No`	FEATURE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)
10	**defect_type** Defect type (one of 5)	text	0.0%	5	Carriageway (8816) Footway (6323) Street lighting (5239) Drainage (5177) Vegetation (5014)	`Street light fault | Carriageway | Flooding or blocked drain`	FEATURE	mapped to 5 canonical values via DEFECT_TYPE_MAPPING (15 variants)
11	**linked_defect_type** Second defect type at the same spot	text	0.0%	6	No linked defect (26659) Drainage (1607) Footway (742) Carriageway (572) Vegetation (565) Street lighting (424)	`Overhanging tree or hedge | No linked defect | None`	FEATURE	mapped to 6 canonical values via LINKED_DEFECT_TYPE_MAPPING (18 variants)
12	**photograph_attached** Photograph attached	text	33.29%	2	Yes (14872) No (5522)	`No | true | Yes`	FEATURE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)
13	**defect_depth_mm** Depth or vertical step, mm	decimal	49.11%	133	5.0 – 165.0 median 40.0	`69 | 50 | 65`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
14	**defect_width_mm** Width across, mm	decimal	49.74%	1,064	51.0 – 3048.0 median 356.0	`280 | 430 | 957`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
15	**water_depth_mm** Depth of standing water, mm	decimal	79.42%	237	13.0 – 420.0 median 80.0	`7 cm | 95 | 18 cm`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
16	**water_location** Where the water stands	text	78.09%	2	Carriageway (4864) Footway (1833)	`Road | Carriageway | Pavement`	FEATURE	mapped to 2 canonical values via HIGHWAY_PART_MAPPING (5 variants)
17	**water_reach_m** Reach of water from the kerb, m	decimal	85.02%	52	0.2 – 5.6 median 1.5	`3.9 m | 1.6 | 2 m`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
18	**overhang_location** Where vegetation overhangs	text	81.97%	2	Footway (4274) Carriageway (1238)	`Pavement | Footway | PATH`	FEATURE	mapped to 2 canonical values via HIGHWAY_PART_MAPPING (5 variants)

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
19	**clear_height_m** Clear height under the overhang, m	decimal	83.3%	52	1.2 – 6.4 median 2.1	`2 m | 2.5 m | 5 ft`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
20	**on_highway_land** Tree / hedge on highway land	text	83.7%	2	Yes (4069) No (915)	`Yes | true | Y`	FEATURE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)
21	**lighting_fault** Street lighting fault	text	81.58%	3	Lamp out (3515) Day burning (1321) Leaning column (794)	`Light not working | DAY | Leaning column`	FEATURE	mapped to 3 canonical values via LIGHTING_FAULT_MAPPING (9 variants)
22	**column_number_given** Lamp column number given	text	81.97%	2	Yes (4430) No (1081)	`false | Y | No`	FEATURE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)
23	**column_lean_degrees** Column lean from upright, degrees	decimal	98.23%	99	1.5 – 22.0 median 5.9	`7 | 11.3 deg | 3 deg`	FEATURE	DEGREE_READING: units degrees → rounded to 0.1; below 0 → blank; above 90 → blank
24	**column_height_m** Column height, m	decimal	81.96%	10	4.9 – 12.0 median 7.9	`5 m | 39 ft | 6`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
25	**previous_reports_at_location** Earlier reports at the location	decimal	0.89%	8	0.0 – 7.0 median 0.0	`0 | 1 | 2`	FEATURE	whole number via SIGNED_WHOLE_NUMBER_PATTERN, Int64; below 0 → blank
26	**planned_crew_hours** Crew hours planned	decimal	1.13%	22	1.0 – 11.5 median 2.5	`120 min | 3.5 | 90 min`	FEATURE	CREW_HOURS_READING: units hours, minutes → rounded to 0.1; below 0.1 → blank
27	**live_street_works_on_street** Live street works permits on the street	whole number	0.0%	5	0 – 4 median 0.0	`0 | 1 | 3`	FEATURE	whole number via WHOLE_NUMBER_PATTERN, Int64
28	**conflicting_street_works** Live permits on the same highway part	whole number	0.0%	4	0 – 3 median 0.0	`0 | 1 | 2`	FEATURE	whole number via WHOLE_NUMBER_PATTERN, Int64
29	**review_outcome** What happened after the decision	text	0.0%	4	Repaired as reported (9680) Repaired after coordination (8514) Repaired after inspection (7411) Closed after inspection (4964)	`Repaired – inspected | Repaired after coordination | Repaired`	LEAKAGE	mapped to 4 canonical values via REVIEW_OUTCOME_MAPPING (12 variants)
30	**reported_again** Defect reported again after repair	text	0.0%	2	No (24393) Yes (6176)	`false | No | Yes`	TARGET	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
31	**repeat_cause** Why it came back	text	79.8%	6	Worsened after inspection (1527) Repair failed (1292) Defect returned nearby (1197) Blocked again (1183) Regrowth (852) Lamp failed again (125)	`Regrowth | Repair failed | Got worse`	LEAKAGE	mapped to 6 canonical values via REPEAT_CAUSE_MAPPING (15 variants)

CSV SCHEMA · data/report_history_eval.csv

Report history · evaluation

Held-out decided reports, same layout as train; never fitted on.

8,202 → 7,648	554	31	reported_again	report_reference (+ revision_number before cleaning)
rows raw → clean	removed: 93 duplicates, 408 earlier revisions, 53 no target	columns	target	primary key

Cleaned file: `processed/report_history_eval_clean.csv` · **Used for:** Scores the repeat likelihood model; chat street figures.

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
1	**report_reference** Report identifier, canonical SC + 7 digits	text	0.0%	7,648	7,648 distinct	`SC0044232 | SC-45339 | SC0044982`	KEY	converted to canonical SC + 7 digits (REPORT_REFERENCE_PATTERN: SC0052125 / SC-52125 / 52125)
2	**report_channel** How the report came in	text	0.0%	3	Website (3193) StreetCare app (2885) Contact centre (1570)	`Website | StreetCare app | Contact centre`	context	mapped to 3 canonical values via REPORT_CHANNEL_MAPPING (3 variants)
3	**revision_number** Revision of the report; only the latest is kept	whole number	0.0%	3	1 – 3 median 1.0	`1 | 2 | 3`	context	whole number via POSITIVE_WHOLE_NUMBER_PATTERN, Int64; only the highest revision of each report is kept
4	**reporter_type** Who reported it	text	0.81%	5	Resident (6279) Business (517) Councillor (377) School (257) Bus operator (156)	`Business | Resident | School`	context	mapped to 5 canonical values via REPORTER_TYPE_MAPPING (5 variants)
5	**street_name** Street of the defect (one of 48)	text	0.0%	48	48 distinct	`Victoria Road | London Road | High Street`	context	mapped to 48 canonical values via STREET_NAME_MAPPING (48 variants)
6	**district** Inspection district (one of 4)	text	0.0%	4	Town Centre (2186) North (2103) South (1929) Villages (1430)	`South | Town Centre | Villages`	context	mapped to 4 canonical values via DISTRICT_MAPPING (4 variants)
7	**street_category** Street category (e.g. Main route, Local road)	text	1.18%	2	Local road (4134) Traffic sensitive (3424)	`traffic sensitive | Main route | Traffic sensitive`	FEATURE	mapped to 2 canonical values via STREET_CATEGORY_MAPPING (6 variants)

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
8	**location_description** Free-text location	text	0.0%	479	479 distinct	`along the street | outside no. 69 | outside no. 11`	context	trimmed; null tokens → blank; otherwise kept as written
9	**location_fixed** Location fixed to a house no., lamp column or junction	text	1.07%	2	Yes (6654) No (912)	`No | true | Yes`	FEATURE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)
10	**defect_type** Defect type (one of 5)	text	0.0%	5	Carriageway (2166) Footway (1550) Street lighting (1358) Drainage (1298) Vegetation (1276)	`CARRIAGEWAY | Pavement defect | Carriageway`	FEATURE	mapped to 5 canonical values via DEFECT_TYPE_MAPPING (15 variants)
11	**linked_defect_type** Second defect type at the same spot	text	0.0%	6	No linked defect (6697) Drainage (384) Footway (182) Vegetation (151) Carriageway (138) Street lighting (96)	`No linked defect | Overhanging tree or hedge | None`	FEATURE	mapped to 6 canonical values via LINKED_DEFECT_TYPE_MAPPING (18 variants)
12	**photograph_attached** Photograph attached	text	34.13%	2	Yes (3639) No (1399)	`Yes | true | No`	FEATURE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)
13	**defect_depth_mm** Depth or vertical step, mm	decimal	49.97%	117	9.0 – 175.0 median 40.0	`70 | 2.5 cm | 40`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
14	**defect_width_mm** Width across, mm	decimal	51.09%	679	50.0 – 4835.0 median 350.0	`25.8 cm | 36 cm | 210`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
15	**water_depth_mm** Depth of standing water, mm	decimal	79.52%	180	15.0 – 365.0 median 81.0	`78 | 80 | 95`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
16	**water_location** Where the water stands	text	78.27%	2	Carriageway (1234) Footway (428)	`Footway | Carriageway | ROAD`	FEATURE	mapped to 2 canonical values via HIGHWAY_PART_MAPPING (5 variants)
17	**water_reach_m** Reach of water from the kerb, m	decimal	85.08%	47	0.3 – 4.9 median 1.5	`2.9 | 6 ft | 1.8`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
18	**overhang_location** Where vegetation overhangs	text	81.56%	2	Footway (1064) Carriageway (346)	`Pavement | Footway | Carriageway`	FEATURE	mapped to 2 canonical values via HIGHWAY_PART_MAPPING (5 variants)
19	**clear_height_m** Clear height under the overhang, m	decimal	83.11%	47	1.3 – 6.1 median 2.2	`1.9 m | 2.1 | 4.3`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
20	**on_highway_land** Tree / hedge on highway land	text	83.16%	2	Yes (1059) No (229)	`True | No | Yes`	FEATURE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)
21	**lighting_fault** Street lighting fault	text	81.09%	3	Lamp out (902) Day burning (316) Leaning column (228)	`Lamp out | Light on in daytime | Light not working`	FEATURE	mapped to 3 canonical values via LIGHTING_FAULT_MAPPING (9 variants)
22	**column_number_given** Lamp column number given	text	81.46%	2	Yes (1151) No (267)	`No | true | N`	FEATURE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)
23	**column_lean_degrees** Column lean from upright, degrees	decimal	98.04%	58	2.0 – 18.5 median 5.55	`3.5 | 7.6 deg | 3 deg`	FEATURE	DEGREE_READING: units degrees → rounded to 0.1; below 0 → blank; above 90 → blank
24	**column_height_m** Column height, m	decimal	81.54%	10	4.9 – 12.0 median 8.0	`5 | 8 | 10 m`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
25	**previous_reports_at_location** Earlier reports at the location	decimal	0.89%	7	0.0 – 6.0 median 0.0	`0 | 2 | 1`	FEATURE	whole number via SIGNED_WHOLE_NUMBER_PATTERN, Int64; below 0 → blank
26	**planned_crew_hours** Crew hours planned	decimal	1.29%	22	1.0 – 11.5 median 2.5	`3.5 | 210 min | 3`	FEATURE	CREW_HOURS_READING: units hours, minutes → rounded to 0.1; below 0.1 → blank
27	**live_street_works_on_street** Live street works permits on the street	whole number	0.0%	5	0 – 4 median 0.0	`1 | 0 | 2`	FEATURE	whole number via WHOLE_NUMBER_PATTERN, Int64
28	**conflicting_street_works** Live permits on the same highway part	whole number	0.0%	4	0 – 3 median 0.0	`1 | 0 | 2`	FEATURE	whole number via WHOLE_NUMBER_PATTERN, Int64
29	**review_outcome** What happened after the decision	text	0.0%	4	Repaired as reported (2434) Repaired after coordination (2056) Repaired after inspection (1867) Closed after inspection (1291)	`Repaired after coordination | Repaired | Closed after inspection`	LEAKAGE	mapped to 4 canonical values via REVIEW_OUTCOME_MAPPING (12 variants)
30	**reported_again** Defect reported again after repair	text	0.0%	2	No (6092) Yes (1556)	`no | false | No`	TARGET	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
31	**repeat_cause** Why it came back	text	79.65%	6	Worsened after inspection (444) Repair failed (335) Blocked again (280) Defect returned nearby (262) Regrowth (207) Lamp failed again (28)	`FAILED | Worsened after inspection | Defect returned nearby`	LEAKAGE	mapped to 6 canonical values via REPEAT_CAUSE_MAPPING (15 variants)

CSV SCHEMA · data/repair_logs_train.csv

Repair logs · train

One row per repaired report, with the crew hours it really took.

26,095 → 25,537
rows raw → clean

558
removed: 303 duplicates, 46 impossible target, 209 no target

27
columns

actual_crew_hours
target

report_reference (one log per report; a repeat is refused)
primary key

Cleaned file: `processed/repair_logs_train_clean.csv` · **Used for:** Trains the expected crew hours model.

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
1	**report_reference** Report identifier, canonical SC + 7 digits	text	0.0%	25,537	25,537 distinct	`SC0017980 | SC0026591 | SC0018532`	KEY	converted to canonical SC + 7 digits (REPORT_REFERENCE_PATTERN: SC0052125 / SC-52125 / 52125)
2	**report_channel** How the report came in	text	0.0%	3	Website (10796) StreetCare app (9641) Contact centre (5100)	`Website | StreetCare app | Contact centre`	context	mapped to 3 canonical values via REPORT_CHANNEL_MAPPING (3 variants)
3	**street_name** Street of the defect (one of 48)	text	0.0%	48	48 distinct	`Oakfield Drive | Victoria Road | Elm Terrace`	context	mapped to 48 canonical values via STREET_NAME_MAPPING (48 variants)
4	**district** Inspection district (one of 4)	text	0.0%	4	Town Centre (7633) North (6786) South (6746) Villages (4372)	`North | South | Town Centre`	context	mapped to 4 canonical values via DISTRICT_MAPPING (4 variants)
5	**street_category** Street category (e.g. Main route, Local road)	text	1.39%	2	Local road (13542) Traffic sensitive (11639)	`Local road | Traffic sensitive | Main route`	FEATURE	mapped to 2 canonical values via STREET_CATEGORY_MAPPING (6 variants)
6	**defect_type** Defect type (one of 5)	text	0.0%	5	Carriageway (7165) Footway (5243) Street lighting (4989) Drainage (4225) Vegetation (3915)	`Street lighting | Footway | Street light fault`	FEATURE	mapped to 5 canonical values via DEFECT_TYPE_MAPPING (15 variants)

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
7	**linked_defect_type** Second defect type at the same spot	text	0.0%	6	No linked defect (21800) Drainage (1522) Footway (705) Carriageway (548) Vegetation (543) Street lighting (419)	`No linked defect | None | Vegetation`	FEATURE	mapped to 6 canonical values via LINKED_DEFECT_TYPE_MAPPING (18 variants)
8	**defect_depth_mm** Depth or vertical step, mm	decimal	49.14%	129	8.0 – 165.0 median 44.0	`20 | 30 | 24`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
9	**defect_width_mm** Width across, mm	decimal	49.77%	1,015	66.0 – 3048.0 median 370.0	`290 | 280 | 140`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
10	**water_depth_mm** Depth of standing water, mm	decimal	79.05%	232	20.0 – 420.0 median 88.0	`2 in | 80 | 6 cm`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
11	**water_location** Where the water stands	text	77.77%	2	Carriageway (4042) Footway (1635)	`ROAD | Carriageway | Footway`	FEATURE	mapped to 2 canonical values via HIGHWAY_PART_MAPPING (5 variants)
12	**water_reach_m** Reach of water from the kerb, m	decimal	85.07%	51	0.3 – 5.6 median 1.6	`3.5 ft | 1.4 | 1.5 m`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
13	**overhang_location** Where vegetation overhangs	text	82.75%	2	Footway (3478) Carriageway (927)	`Footway | Pavement | ROAD`	FEATURE	mapped to 2 canonical values via HIGHWAY_PART_MAPPING (5 variants)
14	**clear_height_m** Clear height under the overhang, m	decimal	83.93%	44	1.2 – 5.7 median 2.1	`2.2 | 1.7 | 2.2 m`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
15	**lighting_fault** Street lighting fault	text	78.95%	3	Lamp out (3505) Day burning (1318) Leaning column (553)	`Lamp out | Light on in daytime | Light not working`	FEATURE	mapped to 3 canonical values via LIGHTING_FAULT_MAPPING (9 variants)
16	**column_height_m** Column height, m	decimal	79.38%	10	4.9 – 12.0 median 8.0	`0 | 8 m | 10`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
17	**column_lean_degrees** Column lean from upright, degrees	decimal	98.51%	87	2.5 – 22.0 median 7.0	`9.8 deg | 6.9 deg | 5`	FEATURE	DEGREE_READING: units degrees → rounded to 0.1; below 0 → blank; above 90 → blank
18	**previous_reports_at_location** Earlier reports at the location	decimal	0.92%	8	0.0 – 7.0 median 0.0	`1 | 0 | 2`	FEATURE	whole number via SIGNED_WHOLE_NUMBER_PATTERN, Int64; below 0 → blank
19	**live_street_works_on_street** Live street works permits on the street	whole number	0.0%	5	0 – 4 median 0.0	`0 | 1 | 4`	FEATURE	whole number via WHOLE_NUMBER_PATTERN, Int64

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
20	**conflicting_street_works** Live permits on the same highway part	whole number	0.0%	4	0 – 3 median 0.0	`0 | 1 | 2`	FEATURE	whole number via WHOLE_NUMBER_PATTERN, Int64
21	**repair_method** How it was repaired	text	0.0%	9	Patch repair (7134) Cut back (3915) Gully cleanse (3385) Lamp replacement (3252) Slab relay (2945) Permanent reinstatement (2329)	`Lamp replacement | Slab relay | Replace photocell`	LEAKAGE	mapped to 9 canonical values via REPAIR_METHOD_MAPPING (25 variants)
22	**traffic_management** Traffic management used	text	0.0%	4	Signs only (9193) None (8652) Lane closure (7611) Road closure (81)	`None | Signs only | No traffic management`	LEAKAGE	mapped to 4 canonical values via TRAFFIC_MANAGEMENT_MAPPING (10 variants)
23	**crew_size** Operatives in the crew	decimal	0.55%	5	1.0 – 5.0 median 2.0	`1 | 2 | 3`	LEAKAGE	whole number via WHOLE_NUMBER_PATTERN, Int64; below 1 → blank
24	**planned_crew_hours** Crew hours planned	decimal	0.93%	21	1.0 – 11.5 median 2.5	`1 | 2.5 | 2`	FEATURE	CREW_HOURS_READING: units hours, minutes → rounded to 0.1; below 0.1 → blank
25	**reported_again** Defect reported again after repair	text	0.0%	2	No (21160) Yes (4377)	`No | NO | false`	LEAKAGE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)
26	**completion_code** How the job completed	text	0.0%	3	Completed (16052) Completed with overrun (7581) Completed after return visit (1904)	`Completed | Done | Completed with overrun`	LEAKAGE	mapped to 3 canonical values via COMPLETION_CODE_MAPPING (9 variants)
27	**actual_crew_hours** Crew hours actually taken	decimal	0.0%	191	0.3 – 43.9 median 2.9	`0.8 | 2.9 | 2.5`	TARGET	CREW_HOURS_READING: units hours, minutes → rounded to 0.1; below 0.1 → blank

CSV SCHEMA · data/repair_logs_eval.csv

Repair logs · evaluation

Held-out repair logs, same layout as train.

6,476 → 6,337	139	27	actual_crew_hours	report_reference
rows raw → clean	removed: 74 duplicates, 16 impossible target, 49 no target	columns	target	primary key

Cleaned file: `processed/repair_logs_eval_clean.csv` · **Used for:** Scores the expected crew hours model.

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
1	**report_reference** Report identifier, canonical SC + 7 digits	text	0.0%	6,337	6,337 distinct	`SC-44085 | SC-44887 | SC-45152`	KEY	converted to canonical SC + 7 digits (REPORT_REFERENCE_PATTERN: SC0052125 / SC-52125 / 52125)
2	**report_channel** How the report came in	text	0.0%	3	Website (2636) StreetCare app (2401) Contact centre (1300)	`StreetCare app | Website | Contact centre`	context	mapped to 3 canonical values via REPORT_CHANNEL_MAPPING (3 variants)
3	**street_name** Street of the defect (one of 48)	text	0.0%	48	48 distinct	`School Lane | Top Road | Riverside Walk`	context	mapped to 48 canonical values via STREET_NAME_MAPPING (48 variants)
4	**district** Inspection district (one of 4)	text	0.0%	4	Town Centre (1831) North (1739) South (1641) Villages (1126)	`Villages | South | North`	context	mapped to 4 canonical values via DISTRICT_MAPPING (4 variants)
5	**street_category** Street category (e.g. Main route, Local road)	text	1.26%	2	Local road (3391) Traffic sensitive (2866)	`Residential street | Main route | Local road`	FEATURE	mapped to 2 canonical values via STREET_CATEGORY_MAPPING (6 variants)
6	**defect_type** Defect type (one of 5)	text	0.0%	5	Carriageway (1717) Footway (1308) Street lighting (1278) Drainage (1050) Vegetation (984)	`Pothole or road surface | Pavement defect | Street light fault`	FEATURE	mapped to 5 canonical values via DEFECT_TYPE_MAPPING (15 variants)

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
7	**linked_defect_type** Second defect type at the same spot	text	0.0%	6	No linked defect (5425) Drainage (359) Footway (181) Vegetation (151) Carriageway (129) Street lighting (92)	`None | No linked defect | Drainage`	FEATURE	mapped to 6 canonical values via LINKED_DEFECT_TYPE_MAPPING (18 variants)
8	**defect_depth_mm** Depth or vertical step, mm	decimal	49.96%	114	10.0 – 175.0 median 44.0	`8 cm | 2.5 cm | 3.7 cm`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
9	**defect_width_mm** Width across, mm	decimal	51.1%	629	75.0 – 4835.0 median 365.0	`33.6 cm | 46.5 cm | 25.6 cm`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
10	**water_depth_mm** Depth of standing water, mm	decimal	79.14%	169	24.0 – 365.0 median 88.0	`170 | 140 | 5.9 cm`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
11	**water_location** Where the water stands	text	78.0%	2	Carriageway (990) Footway (404)	`Carriageway | Road | pavement`	FEATURE	mapped to 2 canonical values via HIGHWAY_PART_MAPPING (5 variants)
12	**water_reach_m** Reach of water from the kerb, m	decimal	85.5%	45	0.3 – 4.9 median 1.6	`1.4 | 1.3 | 1.9 m`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
13	**overhang_location** Where vegetation overhangs	text	82.31%	2	Footway (858) Carriageway (263)	`Road | Pavement | Footway`	FEATURE	mapped to 2 canonical values via HIGHWAY_PART_MAPPING (5 variants)
14	**clear_height_m** Clear height under the overhang, m	decimal	83.67%	40	1.3 – 5.5 median 2.1	`4.5 m | 1.8 m | 2`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
15	**lighting_fault** Street lighting fault	text	78.51%	3	Lamp out (899) Day burning (317) Leaning column (146)	`Light on in daytime | OUT | Light not working`	FEATURE	mapped to 3 canonical values via LIGHTING_FAULT_MAPPING (9 variants)
16	**column_height_m** Column height, m	decimal	78.93%	10	4.9 – 12.0 median 8.0	`5 m | 6 m | 33 ft`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
17	**column_lean_degrees** Column lean from upright, degrees	decimal	98.44%	46	2.6 – 18.5 median 6.9	`7.5 | 5.1 deg | 3.8 deg`	FEATURE	DEGREE_READING: units degrees → rounded to 0.1; below 0 → blank; above 90 → blank
18	**previous_reports_at_location** Earlier reports at the location	decimal	0.9%	7	0.0 – 6.0 median 0.0	`0 | 2 | 1`	FEATURE	whole number via SIGNED_WHOLE_NUMBER_PATTERN, Int64; below 0 → blank
19	**live_street_works_on_street** Live street works permits on the street	whole number	0.0%	5	0 – 4 median 0.0	`0 | 1 | 2`	FEATURE	whole number via WHOLE_NUMBER_PATTERN, Int64

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
20	**conflicting_street_works** Live permits on the same highway part	whole number	0.0%	4	0 – 3 median 0.0	`0 | 2 | 1`	FEATURE	whole number via WHOLE_NUMBER_PATTERN, Int64
21	**repair_method** How it was repaired	text	0.0%	9	Patch repair (1716) Cut back (984) Lamp replacement (851) Gully cleanse (815) Slab relay (757) Permanent reinstatement (552)	`Permanent | Patch | Replace photocell`	LEAKAGE	mapped to 9 canonical values via REPAIR_METHOD_MAPPING (25 variants)
22	**traffic_management** Traffic management used	text	0.0%	4	Signs only (2317) None (2159) Lane closure (1838) Road closure (23)	`Signs | No traffic management | Lane closed`	LEAKAGE	mapped to 4 canonical values via TRAFFIC_MANAGEMENT_MAPPING (10 variants)
23	**crew_size** Operatives in the crew	decimal	0.68%	5	1.0 – 5.0 median 2.0	`4 | 2 | 1`	LEAKAGE	whole number via WHOLE_NUMBER_PATTERN, Int64; below 1 → blank
24	**planned_crew_hours** Crew hours planned	decimal	1.1%	21	1.0 – 11.5 median 2.5	`210 min | 90 min | 60 min`	FEATURE	CREW_HOURS_READING: units hours, minutes → rounded to 0.1; below 0.1 → blank
25	**reported_again** Defect reported again after repair	text	0.0%	2	No (5291) Yes (1046)	`false | true | Yes`	LEAKAGE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)
26	**completion_code** How the job completed	text	0.0%	3	Completed (4082) Completed with overrun (1843) Completed after return visit (412)	`Done | Done – overran | Completed`	LEAKAGE	mapped to 3 canonical values via COMPLETION_CODE_MAPPING (9 variants)
27	**actual_crew_hours** Crew hours actually taken	decimal	0.0%	152	0.3 – 37.7 median 2.9	`198 min | 138 min | 30 min`	TARGET	CREW_HOURS_READING: units hours, minutes → rounded to 0.1; below 0.1 → blank

CSV SCHEMA · data/statement_sentences.csv

Statement sentences

Labelled sentences from documents attached to older reports.

36,661 → 35,717	944	8	sentence_type	sentence_id
rows raw → clean	removed: 445 duplicates, 283 no target, 216 under 3 words	columns	target	primary key

Cleaned file: `processed/statement_sentences_clean.csv` · **Used for:** Trains the sentence type model (stratified 80/20 split, seed 1313).

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
1	**sentence_id** Sentence identifier SEN- + 7 digits	text	0.0%	35,717	35,717 distinct	`SEN–6201357 | SEN–6213400 | SEN–6217536`	KEY	must match SENTENCE_ID_PATTERN (^SEN-[0-9]{7}$); missing, unreadable or repeated → refused (ValueError)
2	**report_reference** Report identifier, canonical SC + 7 digits	text	0.0%	5,902	5,902 distinct	`1196 | SC0003241 | SC–3926`	KEY	converted to canonical SC + 7 digits (REPORT_REFERENCE_PATTERN: SC0052125 / SC-52125 / 52125)
3	**report_channel** How the report came in	text	0.0%	3	Website (15914) StreetCare app (13766) Contact centre (6037)	`Contact centre | Website | StreetCare app`	context	mapped to 3 canonical values via REPORT_CHANNEL_MAPPING (3 variants)
4	**document_type** Kind of document the sentence came from	text	0.0%	4	Resident description (22026) Call note (5101) Photo note (4662) Inspection note (3928)	`Call note | Resident description | Photo note`	context	mapped to 4 canonical values via DOCUMENT_TYPE_MAPPING (4 variants)
5	**sentence_number** Position in its document	whole number	0.0%	7	1 – 7 median 3.0	`1 | 2 | 4`	context	whole number via POSITIVE_WHOLE_NUMBER_PATTERN, Int64
6	**sentence_text** The sentence (cleaned to plain text)	text	0.0%	19,026	19,026 distinct	`Caller reports that a pedestrian was hurt on Station Road. | It is halfway between London`	FEATURE	HTML markup stripped, entities decoded, whitespace collapsed, ALL-CAPS lowered; rows under 3 words dropped
7	**sentence_length_characters** Length of the cleaned sentence	whole number	0.0%	85	16 – 103 median 52.0	`58 | 49 | 61`	context	recomputed from the cleaned text

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
8	**sentence_type** Label: one of 9 sentence types	text	0.0%	9	Measurement (5306) Hazard (5164) Location (4723) Previous reports (3814) Photograph (3768) Weather and water (3521)	`INCIDENT | Location | Weather and water`	TARGET	mapped to 9 canonical values via SENTENCE_TYPE_MAPPING (24 variants)

CSV SCHEMA · data/input/open_defect_reports.csv

Open defect reports (live)

The live queue: reports waiting for review. Same answers as history, without outcomes.

45 → 40	5	27	none	report_reference (+ revision_number before cleaning)
rows raw → clean	removed: 2 duplicates, 3 earlier revisions	columns	target	primary key

Cleaned file: `(cleaned in memory on load)` · **Used for:** The queue, screening, both models at prediction time, the review workflow.

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
1	**report_reference** Report identifier, canonical SC + 7 digits	text	0.0%	40	40 distinct	`SC-52197 | 52163 | 52183`	KEY	converted to canonical SC + 7 digits (REPORT_REFERENCE_PATTERN: SC0052125 / SC-52125 / 52125)
2	**report_channel** How the report came in	text	0.0%	3	Website (14) StreetCare app (13) Contact centre (13)	`StreetCare app | Contact centre | Website`	context	mapped to 3 canonical values via REPORT_CHANNEL_MAPPING (3 variants)
3	**revision_number** Revision of the report; only the latest is kept	whole number	0.0%	2	1 – 2 median 1.0	`1 | 2`	context	whole number via POSITIVE_WHOLE_NUMBER_PATTERN, Int64; only the highest revision of each report is kept
4	**reporter_type** Who reported it	text	0.0%	1	Resident (40)	`Resident`	context	mapped to 5 canonical values via REPORTER_TYPE_MAPPING (5 variants)
5	**street_name** Street of the defect (one of 48)	text	0.0%	40	40 distinct	`Park Avenue | Top Road | Oakfield Drive`	context	mapped to 48 canonical values via STREET_NAME_MAPPING (48 variants)
6	**district** Inspection district (one of 4)	text	0.0%	4	North (12) South (11) Town Centre (9) Villages (8)	`North | Villages | South`	context	mapped to 4 canonical values via DISTRICT_MAPPING (4 variants)
7	**street_category** Street category (e.g. Main route, Local road)	text	2.5%	2	Local road (30) Traffic sensitive (9)	`Main route | LR | Local road`	FEATURE	mapped to 2 canonical values via STREET_CATEGORY_MAPPING (6 variants)
8	**location_description** Free-text location	text	0.0%	38	38 distinct	`outside no. 45 | opposite no. 81 | opposite no. 34`	context	trimmed; null tokens → blank; otherwise kept as written

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
9	**location_fixed** Location fixed to a house no., lamp column or junction	text	2.5%	2	Yes (38) No (1)	`true | Y | Yes`	FEATURE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)
10	**defect_type** Defect type (one of 5)	text	0.0%	5	Carriageway (9) Footway (9) Drainage (8) Street lighting (7) Vegetation (7)	`Flooding or blocked drain | CW | FW`	FEATURE	mapped to 5 canonical values via DEFECT_TYPE_MAPPING (15 variants)
11	**linked_defect_type** Second defect type at the same spot	text	0.0%	6	No linked defect (33) Vegetation (3) Carriageway (1) Street lighting (1) Footway (1) Drainage (1)	`Pothole or road surface | NIL | VG`	FEATURE	mapped to 6 canonical values via LINKED_DEFECT_TYPE_MAPPING (18 variants)
12	**photograph_attached** Photograph attached	text	37.5%	2	Yes (24) No (1)	`true | N | Y`	FEATURE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)
13	**defect_depth_mm** Depth or vertical step, mm	whole number	52.5%	14	19 – 70 median 35.0	`4.5 cm | 2 in | 1.25 in`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
14	**defect_width_mm** Width across, mm	whole number	52.5%	15	121 – 1510 median 450.0	`50 cm | 18 in | 16 in`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
15	**water_depth_mm** Depth of standing water, mm	whole number	80.0%	8	45 – 80 median 57.5	`6 cm | 2 in | 45`	FEATURE	MILLIMETRE_READING: units millimetres, centimetres, inches → rounded to 1; below 1 → blank
16	**water_location** Where the water stands	text	80.0%	2	Carriageway (7) Footway (1)	`Road | ROAD | Carriageway`	FEATURE	mapped to 2 canonical values via HIGHWAY_PART_MAPPING (5 variants)
17	**water_reach_m** Reach of water from the kerb, m	decimal	80.0%	6	0.9 – 2.0 median 1.2	`1.2 m | 3 ft | 1.8`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
18	**overhang_location** Where vegetation overhangs	text	75.0%	2	Footway (7) Carriageway (3)	`PATH | Pavement | ROAD`	FEATURE	mapped to 2 canonical values via HIGHWAY_PART_MAPPING (5 variants)
19	**clear_height_m** Clear height under the overhang, m	decimal	77.5%	8	1.9 – 5.1 median 2.4	`8 ft | 2.5 m | 16.5 ft`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
20	**on_highway_land** Tree / hedge on highway land	text	77.5%	2	Yes (7) No (2)	`Y | true | false`	FEATURE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
21	**lighting_fault** Street lighting fault	text	80.0%	2	Leaning column (4) Lamp out (4)	`LEAN | Lamp out | Column leaning`	FEATURE	mapped to 3 canonical values via LIGHTING_FAULT_MAPPING (9 variants)
22	**column_number_given** Lamp column number given	text	82.5%	2	Yes (6) No (1)	`Yes | No | true`	FEATURE	mapped to 2 canonical values via BOOLEAN_MAPPING (8 variants)
23	**column_lean_degrees** Column lean from upright, degrees	decimal	92.5%	3	3.5 – 7.0 median 5.0	`3.5 deg | 7 deg | 5`	FEATURE	DEGREE_READING: units degrees → rounded to 0.1; below 0 → blank; above 90 → blank
24	**column_height_m** Column height, m	decimal	82.5%	6	4.9 – 10.1 median 6.1	`20 ft | 6 | 6 m`	FEATURE	METRE_READING: units metres, feet → rounded to 0.1; below 0.1 → blank
25	**previous_reports_at_location** Earlier reports at the location	whole number	0.0%	4	0 – 3 median 0.0	`0 | 1 | 2`	FEATURE	whole number via SIGNED_WHOLE_NUMBER_PATTERN, Int64; below 0 → blank
26	**planned_crew_hours** Crew hours planned	decimal	0.0%	9	1.5 – 8.0 median 2.5	`210 min | 2.5 hrs | 2 hrs`	FEATURE	CREW_HOURS_READING: units hours, minutes → rounded to 0.1; below 0.1 → blank
27	**supporting_document_count** Documents attached to the report	whole number	0.0%	2	1 – 2 median 1.5	`2 | 1`	context	whole number via WHOLE_NUMBER_PATTERN, Int64

CSV SCHEMA · data/input/report_documents.csv

Report documents (live)

Documents attached to the open reports.

62 → 60	2	6	none	document_id; report_reference → open_defect_reports
rows raw → clean	removed: 2 duplicates	columns	target	primary key

Cleaned file: `(cleaned in memory on load)` · **Used for:** Statement evidence: sentence model and disagreement flags.

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
1	**document_id** Document identifier DOC- + 6 digits	text	0.0%	60	60 distinct	`DOC-841268 | DOC-841264 | DOC-841220`	KEY	must match DOCUMENT_ID_PATTERN (^DOC-[0-9]{6}$); missing, unreadable or repeated → refused (ValueError)
2	**report_reference** Report identifier, canonical SC + 7 digits	text	0.0%	40	40 distinct	`SC-52128 | SC-52197 | SC0052176`	KEY	converted to canonical SC + 7 digits (REPORT_REFERENCE_PATTERN: SC0052125 / SC-52125 / 52125)
3	**report_channel** How the report came in	text	0.0%	3	StreetCare app (21) Website (21) Contact centre (18)	`StreetCare app | Website | Contact centre`	context	mapped to 3 canonical values via REPORT_CHANNEL_MAPPING (3 variants)
4	**document_type** Kind of document the sentence came from	text	0.0%	3	Resident description (27) Photo note (20) Call note (13)	`Resident description | Call note | Photo note`	context	mapped to 4 canonical values via DOCUMENT_TYPE_MAPPING (4 variants)
5	**document_title** Title of the document	text	0.0%	60	60 distinct	`Resident description – Vegetation – Long Lane | Resident description – Drainage – Park Ave`	context	trimmed; null tokens → blank; otherwise kept as written
6	**document_text** Full document text (markup removed)	text	0.0%	60	60 distinct	`It is on the south side of Long Lane. There is only 5.1 m clear above the road. Children w`	context	HTML markup stripped, entities decoded, whitespace collapsed

CSV SCHEMA · data/input/street_works_register.csv

Street works register (live)

Street works permits; matched to reports by street.

29 → 28	1	6	none	permit_reference; street_name → open_defect_reports
rows raw → clean	removed: 1 duplicates	columns	target	primary key

Cleaned file: `(cleaned in memory on load)` · **Used for:** Live and conflicting permit counts, the live street works trigger.

#	Column · meaning	Type	Null %	Distinct	Range / values	Raw examples (as written)	Role	Cleaning rule
1	**permit_reference** Permit identifier SWP + 7 digits	text	0.0%	28	28 distinct	`SWP0071240 | SWP0071246 | SWP0071255`	KEY	must match PERMIT_REFERENCE_PATTERN (^SWP[0-9]{7}$); missing, unreadable or repeated → refused (ValueError)
2	**street_name** Street of the defect (one of 48)	text	0.0%	26	26 distinct	`Mill Lane | Victoria Road | STATION ROAD`	context	mapped to 48 canonical values via STREET_NAME_MAPPING (48 variants)
3	**works_location** Highway part the works occupy	text	0.0%	2	Carriageway (20) Footway (8)	`Carriageway | Footway`	context	mapped to 2 canonical values via HIGHWAY_PART_MAPPING (5 variants)
4	**works_promoter** Utility / contractor	text	0.0%	5	Water company (7) Telecoms (7) Gas network (6) Electricity network (5) Council contractor (3)	`Water company | Telecoms | Gas network`	context	mapped to 5 canonical values via WORKS_PROMOTER_MAPPING (5 variants)
5	**works_description** What the works are	text	0.0%	11	11 distinct	`Water main repair | Duct repair | Gas main replacement`	context	trimmed; null tokens → blank; otherwise kept as written
6	**permit_status** Live or Cancelled	text	0.0%	2	Live (23) Cancelled (5)	`Live | Cancelled | live`	context	mapped to 2 canonical values via PERMIT_STATUS_MAPPING (2 variants)